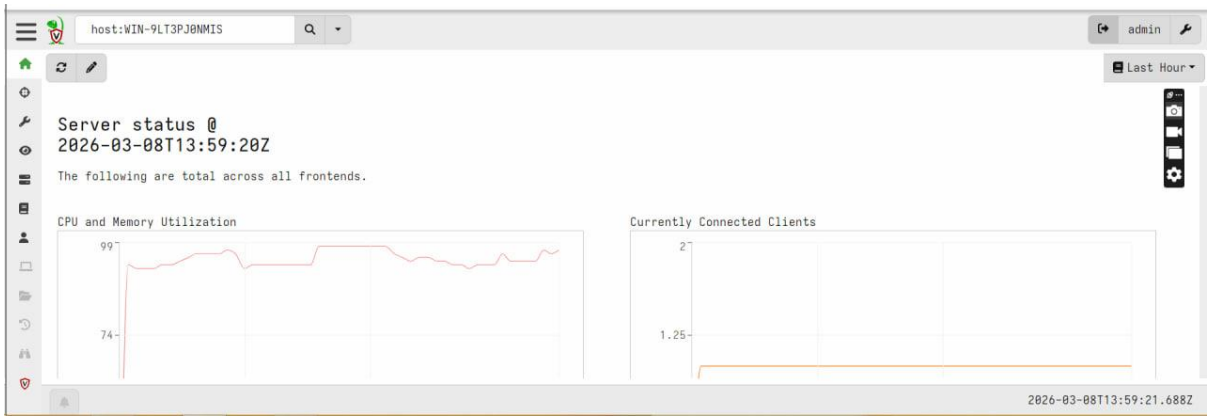
4. Evidence Preservation

By using velociraptor and memory hashing

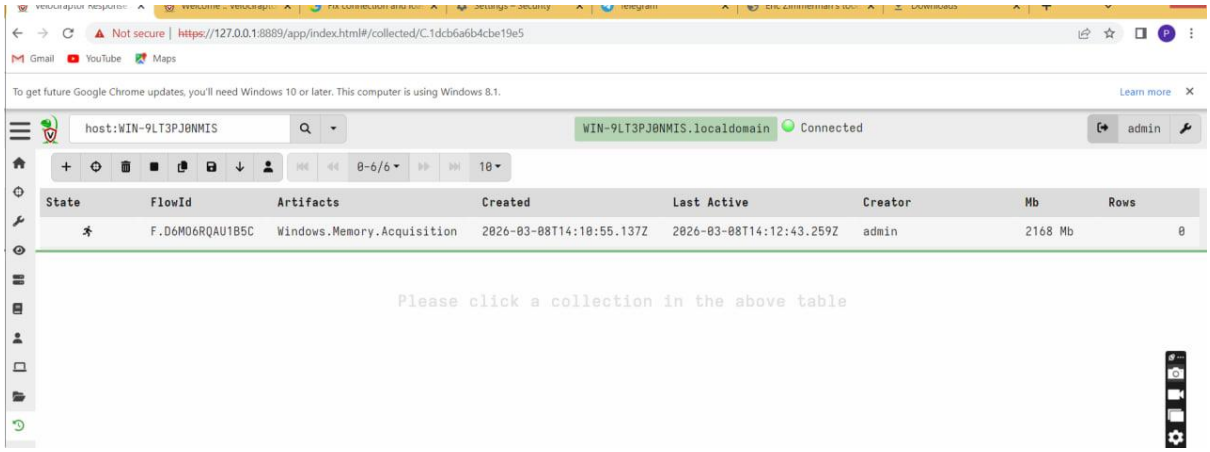
1. Launch Velociraptor

First, I opened the Velociraptor DFIR platform on my system. The Velociraptor web interface was accessed through the browser using the local address https://127.0.0.1:8889.After logging in, the dashboard displayed the connected client machine which was used for evidence collection.



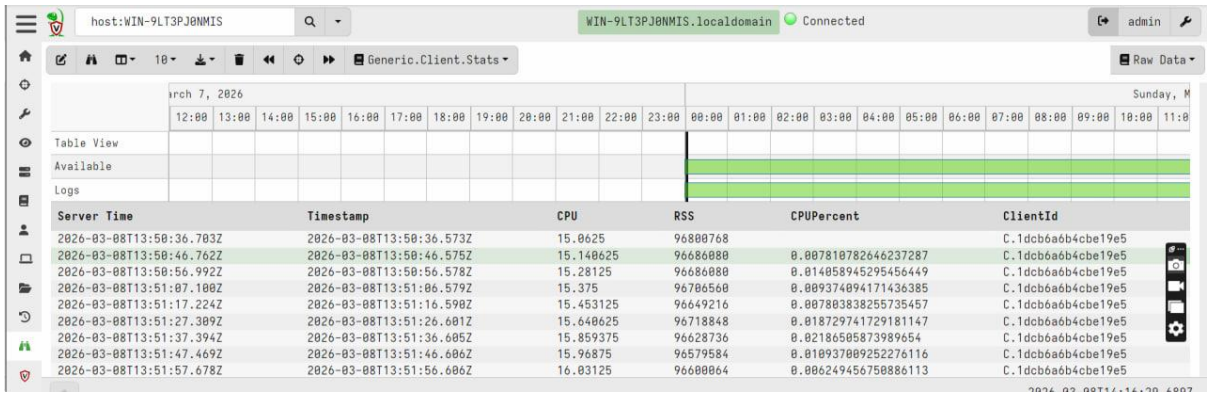
2. Identify the Target System

In the Velociraptor dashboard, the connected Windows virtual machine was identified. The system hostname was visible in the client list and that endpoint was selected for further investigation and evidence collection.



3. Collect Network Connection Evidence

To collect information about active network connections, the artifact Windows.Network.Netstat was executed in Velociraptor. This artifact collects details about running network connections including the local IP address, remote IP address, port numbers, process IDs, program names, and connection state. After running the artifact, Velociraptor displayed all active network connections running on the system.



4. Export Netstat Evidence

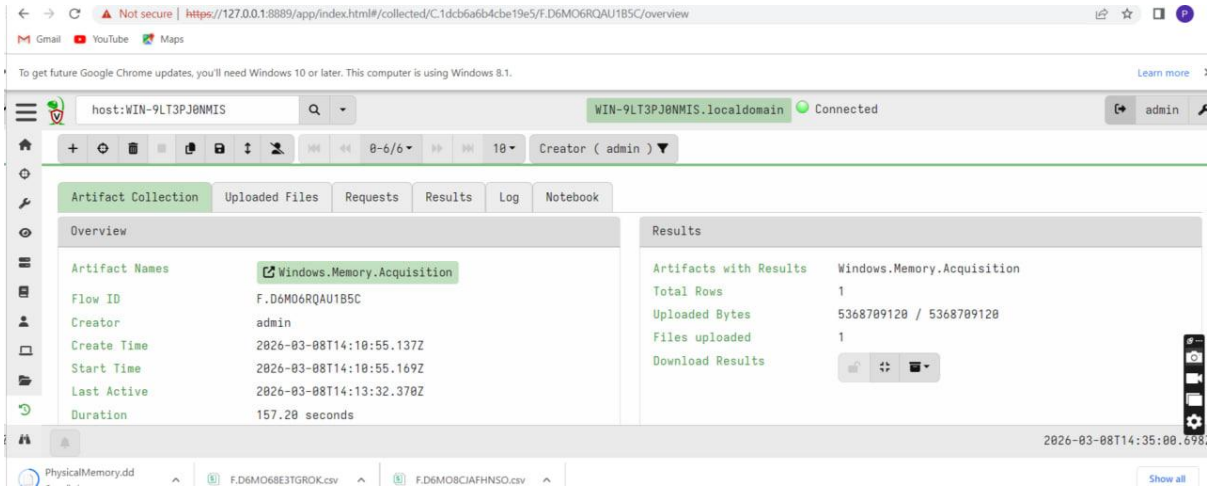
After collecting the network data, the results were exported for documentation. The export format used was CSV.

The exported file contains structured information about network connections which can later be used for forensic investigation.

Pid	Name	Family	Type	Status	Laddr.IP	Laddr.Port	Raddr.IP	Raddr.Port	Timestamp
620	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	135	0.0.0.0	0	1601-01-01T00:00:00Z
4	System	IPv4	TCP	LISTEN	192.168.12	139	0.0.0.0	0	1601-01-01T00:00:00Z
4340	velociraptor	IPv4	TCP	ESTAB	127.0.0.1	8000	127.0.0.1	49365	2026-03-08T13:50:26Z
4340	velociraptor	IPv4	TCP	ESTAB	127.0.0.1	8000	127.0.0.1	49370	2026-03-08T13:50:27Z
4340	velociraptor	IPv4	TCP	LISTEN	127.0.0.1	8000	0.0.0.0	0	1601-01-01T00:00:00Z
4340	velociraptor	IPv4	TCP	ESTAB	127.0.0.1	8000	127.0.0.1	49369	2026-03-08T13:50:27Z
4340	velociraptor	IPv4	TCP	ESTAB	127.0.0.1	8000	127.0.0.1	49463	2026-03-08T14:10:56Z
4340	velociraptor	IPv4	TCP	LISTEN	127.0.0.1	8001	0.0.0.0	0	1601-01-01T00:00:00Z
4340	velociraptor	IPv4	TCP	ESTAB	127.0.0.1	8001	127.0.0.1	49343	2026-03-08T13:50:21Z
4340	velociraptor	IPv4	TCP	LISTEN	127.0.0.1	8003	0.0.0.0	0	1601-01-01T00:00:00Z
4340	velociraptor	IPv4	TCP	ESTAB	127.0.0.1	8889	127.0.0.1	49425	2026-03-08T13:52:54Z
4340	velociraptor	IPv4	TCP	LISTEN	127.0.0.1	8889	0.0.0.0	0	1601-01-01T00:00:00Z
416	wininit.exe	IPv4	TCP	LISTEN	0.0.0.0	49152	0.0.0.0	0	1601-01-01T00:00:00Z
748	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	49153	0.0.0.0	0	1601-01-01T00:00:00Z
796	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	49154	0.0.0.0	0	1601-01-01T00:00:00Z
604	spoolsv.exe	IPv4	TCP	LISTEN	0.0.0.0	49155	0.0.0.0	0	1601-01-01T00:00:00Z
508	services.exe	IPv4	TCP	LISTEN	0.0.0.0	49157	0.0.0.0	0	1601-01-01T00:00:00Z
516	lsass.exe	IPv4	TCP	LISTEN	0.0.0.0	49158	0.0.0.0	0	1601-01-01T00:00:00Z
4048	wpscloudsv	IPv4	TCP	CLOSE_WAI	192.168.12	49204	38.60.148.1	443	1601-01-01T00:00:00Z
4048	wpscloudsv	IPv4	TCP	CLOSE_WAI	192.168.12	49217	38.60.148.1	443	1601-01-01T00:00:00Z
4048	wpscloudsv	IPv4	TCP	CLOSE_WAI	192.168.12	49220	38.60.148.1	443	1601-01-01T00:00:00Z
4048	wpscloudsv	IPv4	TCP	CLOSE_WAI	192.168.12	49238	38.60.148.1	443	1601-01-01T00:00:00Z
4048	wpscloudsv	IPv4	TCP	CLOSE_WAI	192.168.12	49239	38.60.148.1	443	1601-01-01T00:00:00Z
3696	promecefpl	IPv4	TCP	CLOSE_WAI	192.168.12	49244	71.18.1.211	443	1601-01-01T00:00:00Z
3696	promecefpl	IPv4	TCP	CLOSE_WAI	192.168.12	49245	71.18.1.211	443	1601-01-01T00:00:00Z
3696	promecefpl	IPv4	TCP	CLOSE_WAI	192.168.12	49246	71.18.1.211	443	1601-01-01T00:00:00Z
3696	promecefpl	IPv4	TCP	CLOSE_WAI	192.168.12	49247	71.18.1.211	443	1601-01-01T00:00:00Z
3696	promecefpl	IPv4	TCP	CLOSE_WAI	192.168.12	49292	3.108.89.12	80	1601-01-01T00:00:00Z

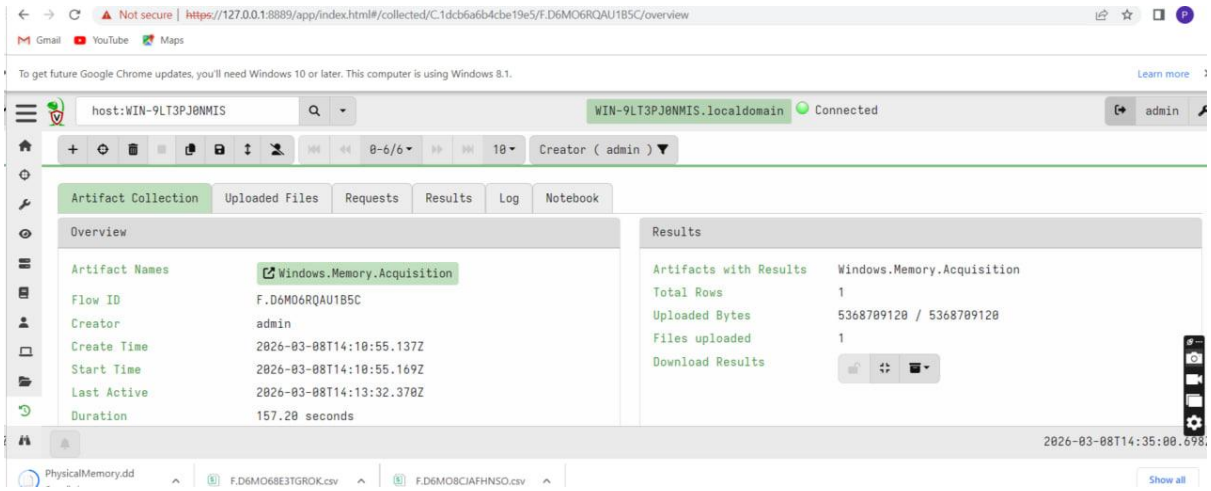
5. Perform Memory Acquisition

To collect volatile memory evidence, the artifact Windows.Memory.Acquisition was executed in Velociraptor. This artifact captures the complete RAM of the system. Memory acquisition is important in digital forensics because RAM may contain running processes, malware artifacts, active network sessions, encryption keys, and suspicious scripts.



6. Memory Dump Collection

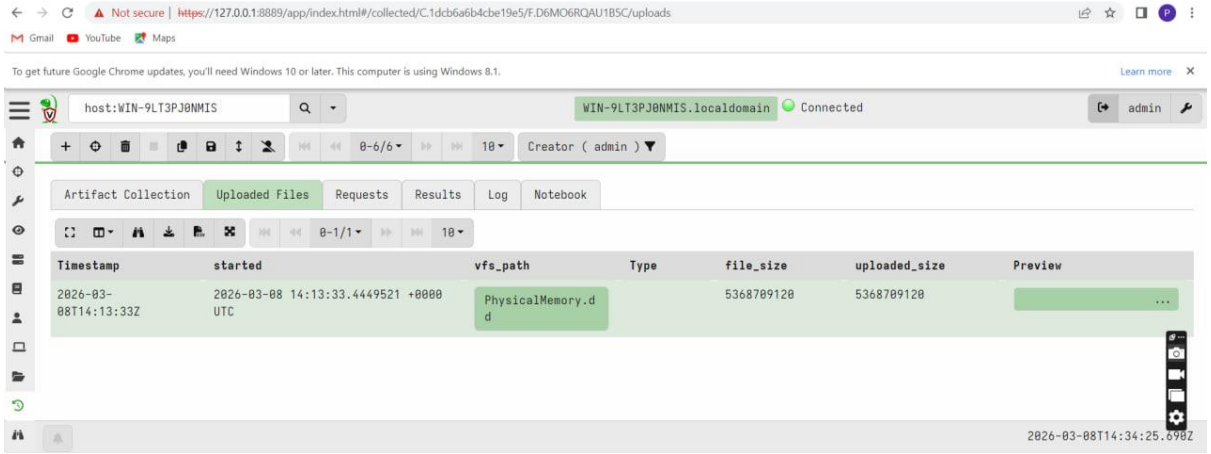
After running the memory acquisition artifact, the system generated a memory dump file named PhysicalMemory.dd. The file size was 5368709120 bytes which is approximately 5 GB. This file represents a full snapshot of the system's RAM at the time of collection.



7. Evidence Integrity Verification

To make sure the collected evidence was not modified, a SHA256 cryptographic hash was generated. The hash was created using the Windows certutil utility by running

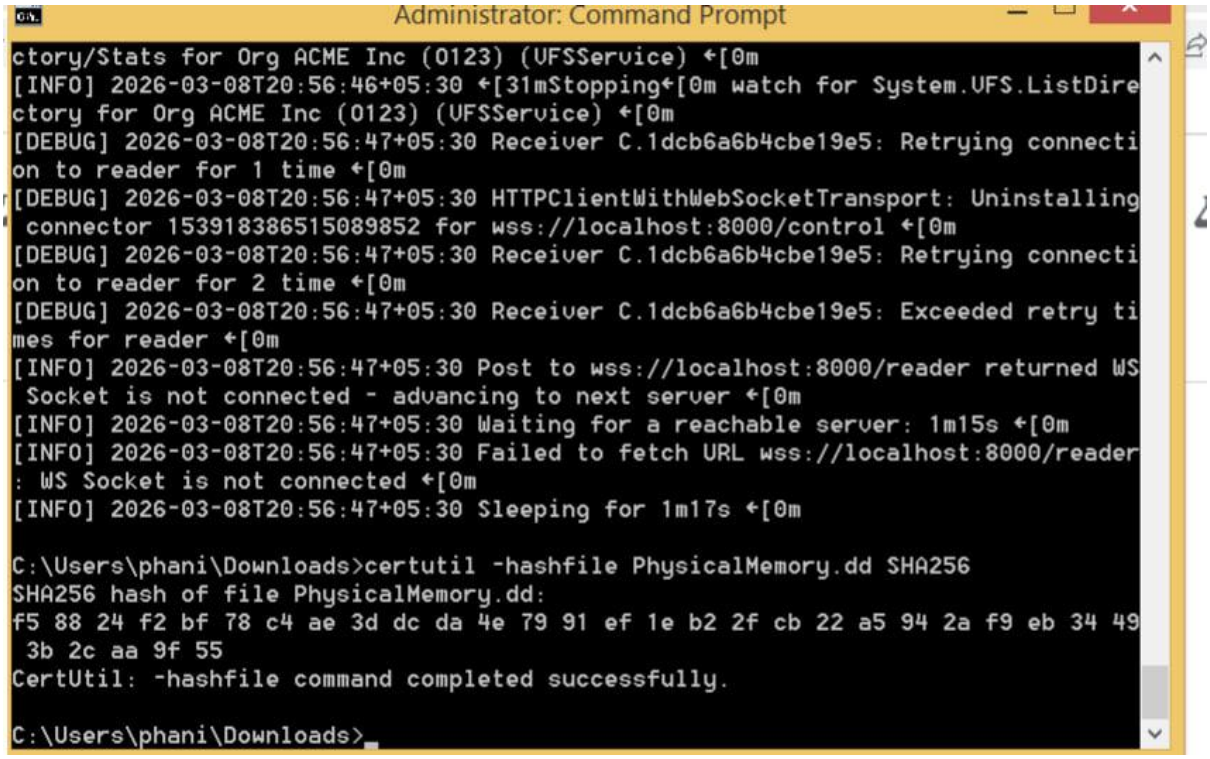
the command to calculate the SHA256 value of the memory dump file.



8. Generated Hash Value

The generated SHA256 hash value of the memory dump file was:
f5882ffb78c4303bdcda4e7991ef1eb2fcbb22a5942af9a031f90a2ceaa0f55

This hash value ensures that the memory dump remains unchanged and preserves the integrity of the digital evidence.



9. Evidence Documentation

All collected artifacts were documented as part of the evidence preservation process.

Item	Description
Network Connections	Netstat CSV file
Memory dump	Physical Memory.dd

Conclusion

In this task, evidence preservation was successfully performed using Velociraptor. Network connection data and a full memory dump were collected from the Windows system. The memory dump was verified using SHA256 hashing to maintain forensic integrity. These artifacts can be used later for further forensic analysis and incident investigation.